

DEWDRIVE · DESIGN PRECIS

DewDrive design precis

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DOCUMENT CONTROL

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REVISION HISTORY

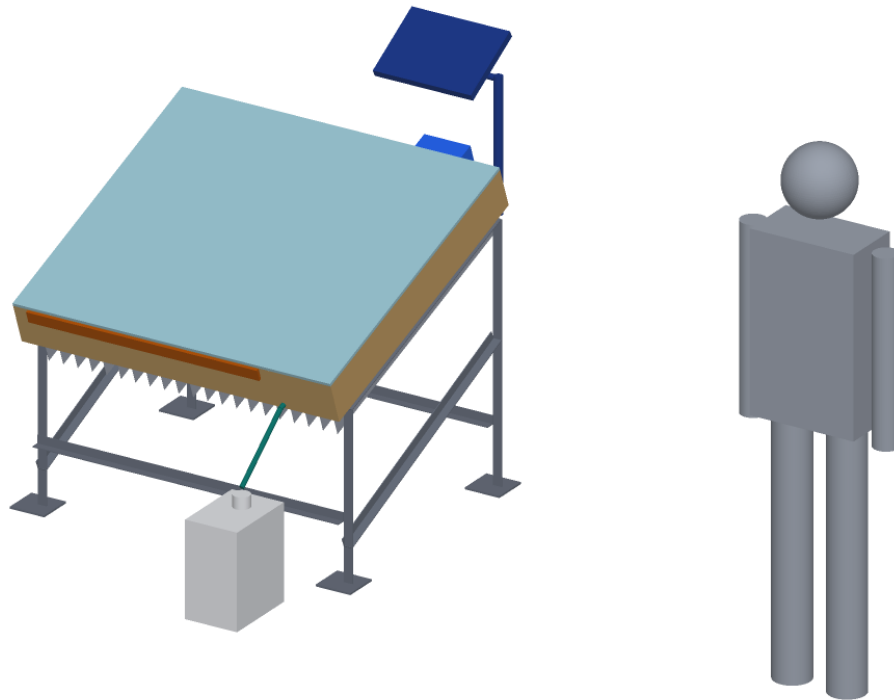
Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); through-flow trays, 25 wt % salt, drip screens, fan humidity cut-out, anchors as default; numbers from DWD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 update (design choices decided by Amish in DWD-DDR-001; numbers replaced by DWD-CAL-001; parametric model and drawing DWD-DWG-001; yield shortfall and proposed fixes)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (how it works, components, first-order numbers, design choices, safety, open questions, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive design precis

DewDrive is a glazed, insulated box 1.1 x 1.0 m, tilted 20° toward the sun on a steel stand. Inside, four black trays hold 4 kg of silica gel impregnated with 25 wt % calcium chloride. At night the vent flaps are opened and a 2 W fan draws air in above the trays and down through their mesh floors, and the sorbent takes up water. In the morning the flaps are shut; the sun heats the bed to about 112 °C by noon and drives the vapour down past a black drip screen onto a finned aluminium floor that forms the condenser, shaded by the box itself. The condensate runs down the slope into a gutter and a 10 L bottle. The TRL 3 calculation (DWD-CAL-001 v0.2) gives about 0.57 L per day at 40 % night RH and 0.45 L at 25 %, meeting the 0.5 and 0.25 L targets. The design follows Amish's decisions of 2026-09-25 (DWD-DDR-002), which replaced the v0.1 layout that made only 0.20 L per day. Parts cost \$515 against the \$520 budget (topped up by Amish on 2026-09-26), which is met. One unit supplements drinking water; it does not replace a water supply.

DewDrive

CONCEPT, NOT FOR FABRICATION



Grey: 1.75 m person for scale

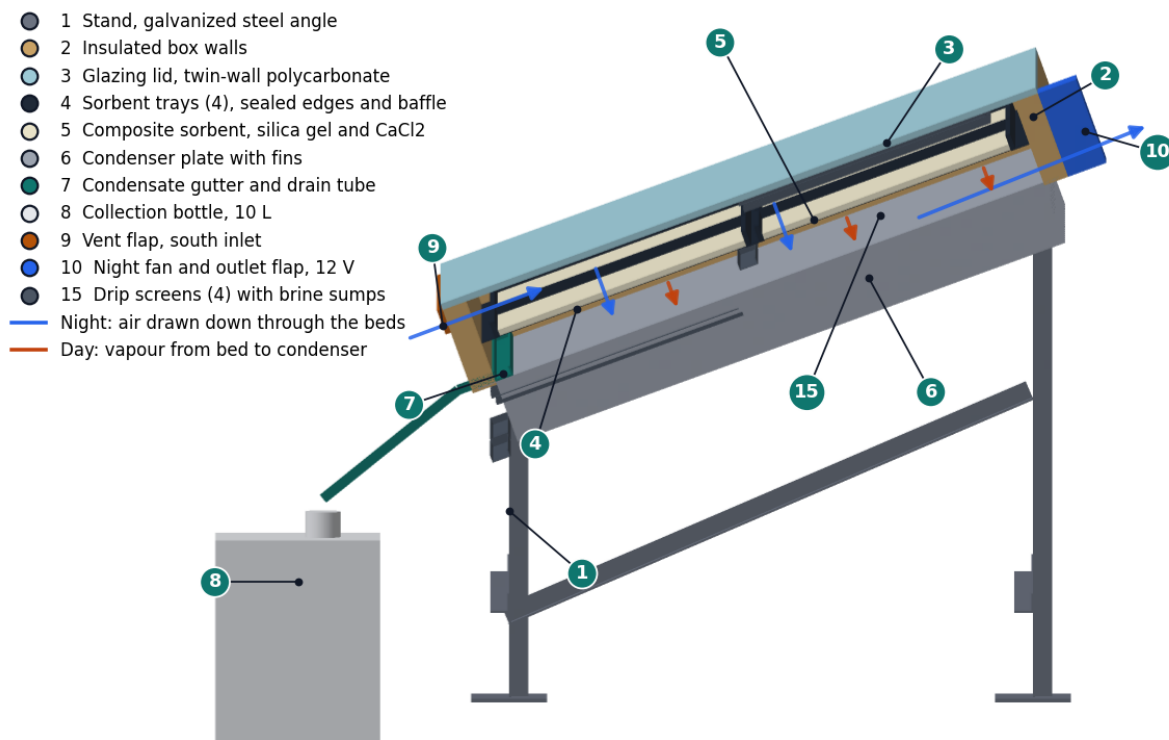
Figure 1. DewDrive on its stand, with a 1.75 m person for scale. Glazing faces south (for a northern-hemisphere site), the collection bottle sits below the low edge and the 10 W PV panel is on a pole to the north, where it cannot shade the box. Generated from the parametric model `cad/src/model.py`.

■ How it works

1. **Adsorb at night.** In the evening the user opens the south inlet flap, which sits above the trays, and the north outlet flap, which sits below them. The logger switches on the fan, which draws 40 m³/h of night air in over the trays, down through the 9.3 mm sorbent beds and their mesh floors, and out under the trays, for 10 h. The trays sit on an EPDM gasket on a sealing baffle, so the air cannot bypass the beds. The CaCl₂ in the silica gel pores binds water first as hydrates and then as a solution held in the pores; the silica gel adds a little uptake of its own and keeps the solution in place. If the air goes above 70 % RH the logger stops the fan.
2. **Seal and heat by day.** In the morning the user shuts both flaps. Sunlight passes through the twin-wall polycarbonate and heats the black top faces of the trays. The bed passes 89 °C by 09:00 and reaches about 112 °C at noon, and its vapour pressure rises above that of the condenser.
3. **Condense.** Vapour diffuses down through the 65 mm gap under the trays, around the drip-screen channels, to the aluminium floor plate, which stays within about 12 K of ambient thanks to 21 fins in the shade under the box.
4. **Collect.** Water films run down the 20° slope to a gutter along the low edge and drain through a silicone tube into a food-grade jerrycan.
5. **Log.** The logger records air temperature and RH, bed and condenser temperatures every 5 min, and applies the fan humidity cut-out; the user weighs the bottle each day. The data make yields comparable between sites.

DewDrive: cutaway (north-south section through the drain, looking west)

CONCEPT, NOT FOR FABRICATION



South (low edge) at left. Flaps open at night; closed by day while the sun heats the bed.

Figure 2. North-south section through the drain, looking west. Blue arrows: night air path with the flaps open, in above the trays and down through the beds. Red arrows: daytime vapour path from the bed (5) past the drip screen (15) to the condenser floor (6). The gutter (7) along the low edge drains to the bottle (8).

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv`, drawing DWD-DWG-001 and Figures 2 and 3.

#	Component	Choice	Notes
1	Stand	Galvanized steel angle 30 x 30 x 3 mm, four legs, bolted; tilt 20°	Separable from the box; two ground anchors by default, about 50 kg of ballast as the alternative (DWD-DDR-002, D13; DWD-CAL-001, H2 to H4)
2	Insulated box walls	Plywood skins with 25 mm PIR foam, 1,100 x 1,000 x 140 mm, 40 mm thick	Open top (glazing) and bottom (condenser)
3	Glazing lid	10 mm UV-stabilized twin-wall polycarbonate, hinged on the north edge	Must be rated above the 122 °C stagnation temperature of the bed, or shown to stay inside its rating (DWD-DDR-002, O5)
4	Sorbent trays (4)	Aluminium pans 490 x 430 x 25 mm with stainless mesh floors; black top, bare underside; EPDM edge gasket on a 1 mm aluminium sealing baffle	Lift out for recharging; the baffle forces the night air through the beds (D9)
5	Composite sorbent	4 kg: 3.0 kg mesoporous silica gel with 1.0 kg CaCl ₂ (25 wt %, D10)	Bed 9.3 mm deep on 0.785 m ² ; all food-grade
15	Drip screens (4)	Two staggered layers of 20 x 6 mm aluminium channels at 30 mm pitch under each tray, running down the slope to a closed brine sump; painted matt black	Catch any brine that leaves the beds (sumps 0.58 L in all) with no line of sight from bed to condenser (D11)
6	Condenser	2 mm aluminium floor plate with 21 fins, 1 x 100 x 920 mm at 50 mm pitch	Wetted face food-safe coated
7	Gutter and drain tube	Gutter along the low edge; food-grade silicone tube	
8	Collection bottle	10 L HDPE jerrycan	About 17 days of production at the design point
9	Vent flap, south inlet	Hinged flap 800 x 40 mm above the trays, with EPDM seal and latches	Closed by day; seal quality sets the vapour loss
10	Night fan and outlet flap	12 V 120 mm fan, about 2 W, on the north wall below the trays	Fan switched by the logger; stops above 70 % RH
11	PV panel	10 W, 12 V, on a pole north of the box	Runs fan and logger
12	Electronics box	PWM charge controller, 12.8 V 6 Ah LiFePO ₄ with BMS, ESP32 logger	Only low-voltage DC on the device
13	Sensors	Air T and RH in a radiation shield; bed and condenser probes	The air RH reading drives the fan cut-out

Item 14 (hardware and consumables) is in the BOM but not modelled. The general arrangement is drawing DWD-DWG-001 Rev P2 (`cad/drawings/`); the STEP files are in `cad/step/` .

Fan rule (DWD-DDR-002, D11). The logger runs the fan from the evening flap opening for 10 h, and stops it whenever the air RH is above 70 % (restarting below about 65 %, to avoid cycling). This is a design rule for the firmware sketch; firmware beyond a sketch is TRL 4 work and on hold.

DewDrive: exploded view

CONCEPT, NOT FOR FABRICATION

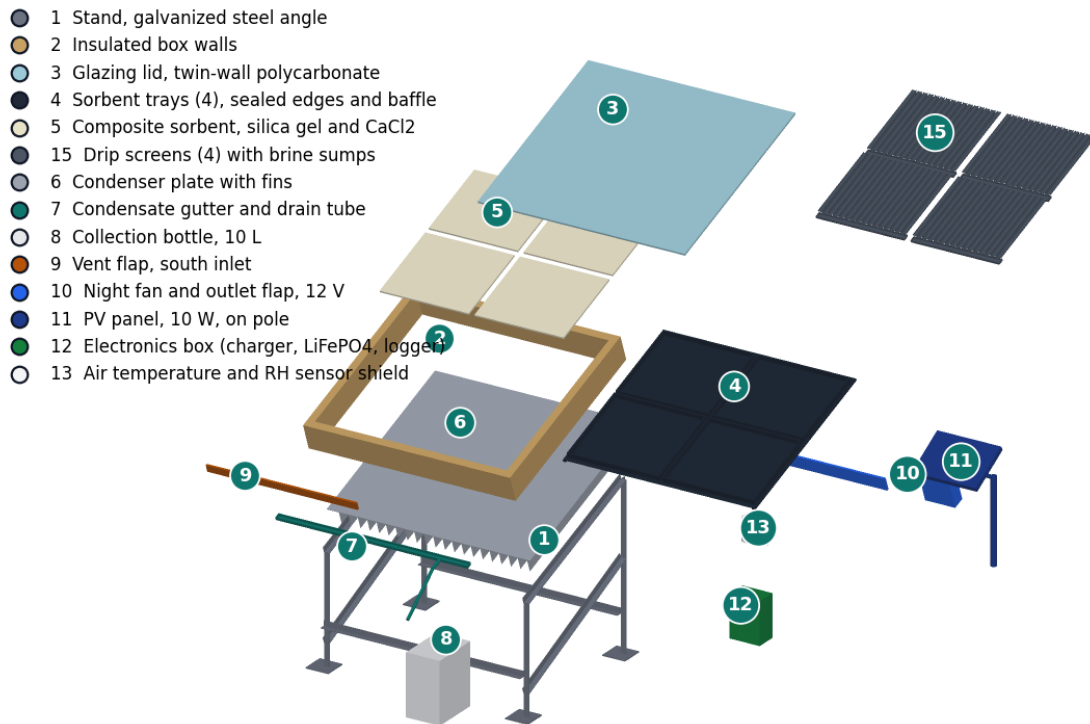


Figure 3. Exploded view with BOM numbers.

Numbers from DWD-CAL-001

All values are first-principles estimates from `docs/04-calcs/sizing.py` (DWD-CAL-001 v0.2); the tag after each value is the line of the script output that carries it. The sorbent isotherm is built from bulk salt data and must be replaced by measured isotherms of the actual composite.

DewDrive: water per day at the design point, g (estimates; 40 % night RH, 20 C)

CONCEPT, NOT FOR FABRICATION

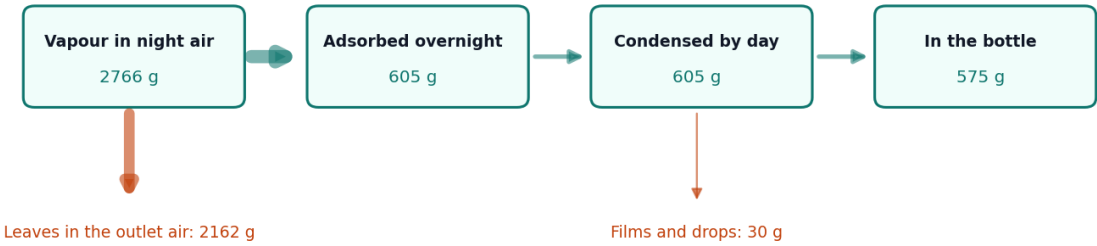


Figure 4. Water per day at the design point (40 % night RH, 20 °C). All values are estimates.

Table 2. Water, energy, size and cost.

Quantity	Value	Basis	Requirement
Vapour carried through the box overnight	2.77 kg (B4)	40 m ³ /h for 10 h at 6.92 g/m ³	
Share of the vapour driving force captured	Nearly all (gas-side NTU 19, B3)	Air drawn down through the bed; 0.16 Pa pressure drop	v0.1 layout: 11 % (B2)
Overnight uptake	0.60 kg (B5)	Cyclic simulation; equilibrium would allow 1.59 kg (A4)	
Collected at 40 % night RH	0.57 L per day (C3)	0.60 kg condensed less 30 g of films	R1 met
Collected at 25 % night RH	0.45 L per day (C4)	Same model	R2 met
Bed temperature at noon	112 °C (C2)	785 W/m ² peak; 4.24 kWh/m ² from 09:00 to 15:00 (C1)	R3 met
Condenser rise above ambient, peak	12.1 K (C2)	Load 299 W peak; UA 24.6 W/K (C8)	R5 met
Where the sun goes	3.68 kWh absorbed; 0.44 kWh desorbs water; 1.69 kWh lost through the glazing; 1.47 kWh to the condenser (C7)	The black drip screens cut the effective bed-to-condenser emissivity from 0.545 to 0.327 (C5)	
Solar-to-water efficiency	7.1 % (C7)	Published devices reach about 9 % (LaPotin et al., 2021)	
Stagnation temperature, dry bed	122 °C (D1)	Noon, 35 °C air	Glazing rating to check (O5)
Electrical use and supply	23.6 Wh per day against 35 Wh (F1); 2.6 days of autonomy (F2)	Fan 2 W for 10 h; logger 0.15 W	R4 met
Salt containment	Pores fill above 71 % RH (E2); 43 % full after one 90 % RH night (E3)	Sumps 0.58 L (E6)	R7 met on paper
Mass	53.9 kg (119 lb) dry; box 33.4 kg, 24.0 kg with trays, sorbent and screens out (G1, G2)	Model volumes	R9 met
Wind at 20 m/s	317 N normal to the lid; tips without two anchors or 46 kg of ballast (H1 to H4)	Force coefficient 1.2 on 1.1 m ²	R10 not verifiable at TRL 3
Parts cost	\$515 (J1)	bom/bom.csv	R11 met (\$520)
Water per dollar over 5 years	about \$0.49 per litre (J2)	Parts only	For comparison

What the numbers say:

- **Drawing the air through the bed fixes the yield.** The v0.1 layout, with air skimming past the trays, captured only 11 % of the vapour driving force and made 0.20 L per day. Through the bed, the air leaves close to equilibrium

with the sorbent, and the day desorption now sets the limit. Even if slow diffusion inside the grains cut the night NTU to 1, the yield would still be 0.53 L per day (C9).

- **The drip screens help the yield.** Black channels under the trays catch brine and also cut the radiation from the hot beads to the condenser by about 40 %: 0.57 L per day with them against 0.51 L without (C6).
- **A low-emissivity screen would do more, but overheats the box.** Bare aluminium channels would give 0.82 L per day, but a dry bed would stagnate at about 166 °C, beyond the rating of polycarbonate glazing (C10, D1). That option is proposed for further paper work (DWD-DDR-002, O4), not adopted.
- **The lower salt loading contains the brine.** At 25 wt % the solution fills the pores only above 71 % RH, against 51 % RH at the v0.1 loading (E2).
- **Cost is within the topped-up budget.** The gasket, baffle and drip screens add \$27; the total is \$515 against the \$520 budget that Amish approved on 2026-09-26 (D15, closing O3).
- **Yield is still modest.** About 0.57 L per day is about one fifth of one person's survival drinking need.

■ Key design choices

Choices 1 to 7 were decided by Amish on 2026-09-25 (DWD-DDR-001, going with the recommendation), and choices 8 to 11 on the same day (DWD-DDR-002). Choice 12 was part of the design from the start.

1. **Single-stage, flat glazed box** rather than a dual-stage or tubular design (D4). Dual-stage could add about 20 % in later versions (LaPotin et al., 2021).
2. **Silica gel and CaCl₂ composite**, rather than MOF, zeolite or LiCl (D3), at 25 wt % salt (D10, amending the 33 wt % of D3).
3. **Condenser as the shaded floor of the box**, directly below the bed (D4). Short vapour path, no pump, one fewer seal.
4. **Fan-assisted night airflow** with a 10 W PV panel and a 12.8 V LiFePO₄ battery, rather than natural airflow (D5).
5. **Manual flaps** opened and closed by the user, rather than automatic actuators (D6).
6. **20° tilt, fixed**, facing the equator, with twin-wall polycarbonate glazing rather than glass (D7).
7. **Design point** of 20 °C and 40 % RH at night and 6.0 kWh/m² of sun, with a 25 % RH low-humidity case (D8).
8. **Night air drawn down through the sealed trays** (D9), with the inlet flap above the trays and the fan below them.
9. **Drip screens and a fan humidity cut-out** for wet spells (D11). The screens are black, not low-emissivity, after the paper evaluation of D12.
10. **Two ground anchors by default**, ballast as the alternative (D13).
11. **Budget kept at \$500** (D14), then topped up to \$520 by Amish on 2026-09-26 (D15) once the BOM reached \$515.
12. **Built-in logging** so that each prototype yields comparable data.

■ Safety

Safety: Harvested water must be tested and treated before drinking. Condensed water is close to distilled, but it can pick up bacteria, dust, salt from the sorbent and metal from the condenser. Until a laboratory test shows otherwise, boil or disinfect it and do not use it as the only drinking source. Distilled-like water also lacks minerals.

Safety: The inside of the box gets hot enough to burn: an estimated 112 °C in the bed at noon and about 122 °C with a dry bed (DWD-CAL-001 v0.2, C2 and D1). The glazing outer skin reaches about 57 °C and the condenser fins about 47 °C. Open the lid only when cool, wear gloves when handling trays and drip screens, and label the glazing. The polycarbonate must be rated above the stagnation temperature (DWD-DDR-002, O5). Do not fit bare, low-emissivity screens: they would raise the dry bed to about 166 °C.

Safety: Calcium chloride irritates eyes and skin and gives off heat when it dissolves. Wear gloves and eye protection when preparing the sorbent and when emptying the drip-screen sumps. At 25 wt % the solution fills the pores only above about 71 % RH (DWD-CAL-001, E2); the fan stops above 70 % RH and the drip screens catch any brine. Keep trays level, close the flaps in rain, and never let the solution reach the water path.

Safety: The 12.8 V LiFePO₄ battery has a built-in BMS and must be fused at the battery. Keep the electronics box shaded, and do not charge a damaged or swollen battery.

Safety: The tilted lid catches the wind like a sail (317 N at 20 m/s). Without restraint the unit tips over (DWD-CAL-001, H2). Fit the two ground anchors, or about 50 kg of ballast, before loading the box. Deburr all aluminium fins, channels and sheet edges; fins under the box are at ankle and hand height.

■ Open questions

- [] What are the measured uptake isotherms and rates of the 25 wt % composite at 20 °C and at 80 to 120 °C? These set every yield number.
- [x] How can the parts cost come back to \$500, or should the budget rise? Decided by Amish, 2026-09-26: budget top-up to \$520 (DWD-DDR-002, D15).
- [] Is a low-emissivity screen worth pursuing with a stagnation vent or a higher-rated glazing (O4, awaiting Amish)?
- [] Does the polycarbonate stay inside its rating with a 122 °C dry bed (O5, awaiting Amish)?
- [] How much salt carries over into the condensate, and does the aluminium condenser need a coating or a stainless replacement?
- [] How well do simple flap and tray seals hold, by night against bypass air and by day against vapour loss?
- [] Which site and partner provide the humidity and solar data for the design point (DWD-DDR-001, O1)?